

# Friction: friend and enemy

## One force. Different jobs.

Explain what friction does, and when it helps or hinders.

FORCES / CONTACT / EVIDENCE



# Arrival • Supported route

FOUR QUICK QUESTIONS / POINT, SAY, SIGN, DRAW OR WRITE / YOU MAY PASS

1. A push moves a toy car away. Which action moved it?  
Push / pull.

2. Look at two palms pressed together. Which surfaces touch? Palm and palm / palm and sleeve.

3. Use your chosen route. What do you notice when hands rub? Say, point or draw one observation.

4. A shoe needs to grip the ground. Does grip help walking or make slipping easier? Help walking / make slipping easier.

For question 3, hand rubbing is optional. If comfortable, gently rub your hands for up to five seconds; stop at discomfort. You can watch an adult or describe a familiar hand-rubbing example instead. Say whether you observed it, remembered it or predicted it.

# Arrival • Standard route

FOUR QUICK QUESTIONS / POINT, SAY, SIGN, DRAW OR WRITE / YOU MAY PASS

1. Name the two actions that can change how an object moves: p\_\_\_ and p\_\_\_.

2. When hands rub, name the two touching surfaces.

3. Use your chosen route. Describe one change you notice or already know. If you know the force's name, add it.

4. A bike brake pad presses on the wheel rim. Predict what happens to the turning wheel.

For question 3, hand rubbing is optional. If comfortable, gently rub your hands for up to five seconds; stop at discomfort. You can watch an adult or describe a familiar hand-rubbing example instead. Say whether you observed it, remembered it or predicted it.

# Arrival • Stretch route

## FOUR QUICK QUESTIONS / POINT, SAY, SIGN, DRAW OR WRITE / YOU MAY PASS

1. A toy car is pushed, then slows on a level floor. Name one force you think could slow it. Not sure is useful.

2. A hand grips a door handle. Name the touching surfaces and explain what the grip helps you do.

3. Use your chosen route. Suggest why rubbing hands may feel warmer. Give a prediction if you cannot observe it.

4. Two toy cars start at different speeds on different surfaces. Is that a fair surface comparison? Name one thing to keep the same.

For question 3, hand rubbing is optional. If comfortable, gently rub your hands for up to five seconds; stop at discomfort. You can watch an adult or describe a familiar hand-rubbing example instead. Say whether you observed it, remembered it or predicted it.

# Three things to work out

1 What is friction?

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2 When does friction help or hinder?

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3 What will happen on two different surfaces?

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# Friction knowledge organiser (1 of 2)

**Objective:** Explain what friction does and give examples where it helps and where it hinders.

## KEY WORDS

Friction — A force that resists sliding, or attempted sliding, between touching surfaces.

Surface — The outside face of an object where contact happens.

Grip — Friction that helps prevent slipping.

Wear — Damage caused by repeated rubbing.

Lubricant — A substance that can reduce friction in suitable contacts.

## SUCCESS CRITERIA

1. Name the two touching surfaces.
2. Explain how friction helps or hinders the intended job.
3. Use an observation or a clearly labelled model to support an idea.

# Friction knowledge organiser (2 of 2)

## KEY IDEAS AND MODEL

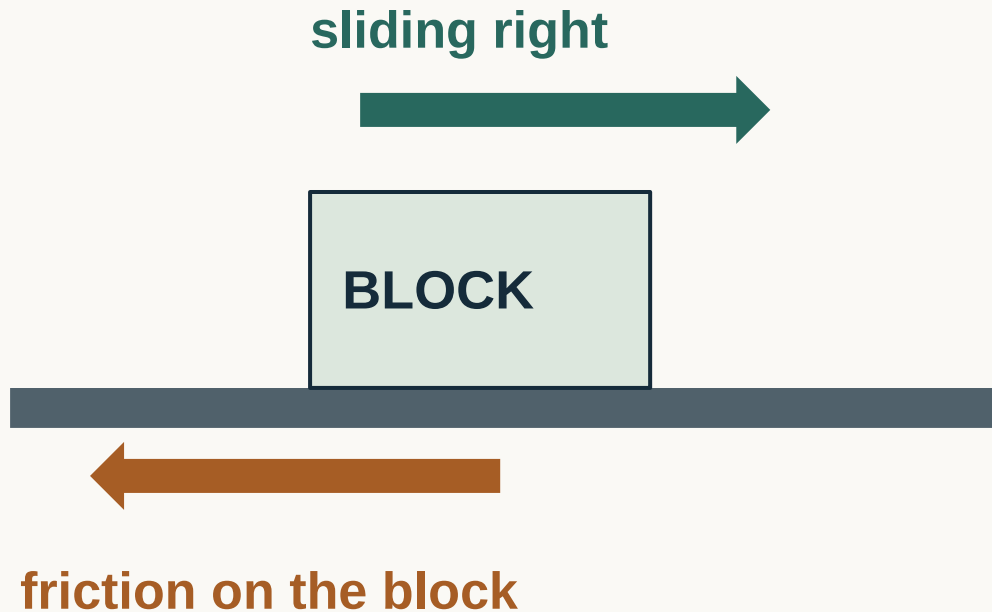
Friction acts at a contact. Identify both surfaces and their relative sliding.  
The actual surface pair and its condition matter. Smooth rubber and smooth glass can still grip; roughness alone does not predict every result.  
Rubbing can transfer energy to thermal stores. A brief rub need not produce noticeable warmth.  
Model: For a box sliding right on a floor, friction on the box acts left along the contact.  
This is an explanatory model, not a force measurement.

## HELPFUL OR UNHELPFUL?

Helpful: pad-wheel friction helps a bike brake slow the wheel.  
Unhelpful: rubbing between moving engine parts can cause unwanted heating and wear.

Use evidence: Compare surfaces using the same block, fixed ramp and release point, no push, and the same measuring start. Repeat a run. Record what actually happened; label supplied results as model data.

# Friction acts at a contact



A force between surfaces that slide, or try to slide, past each other.

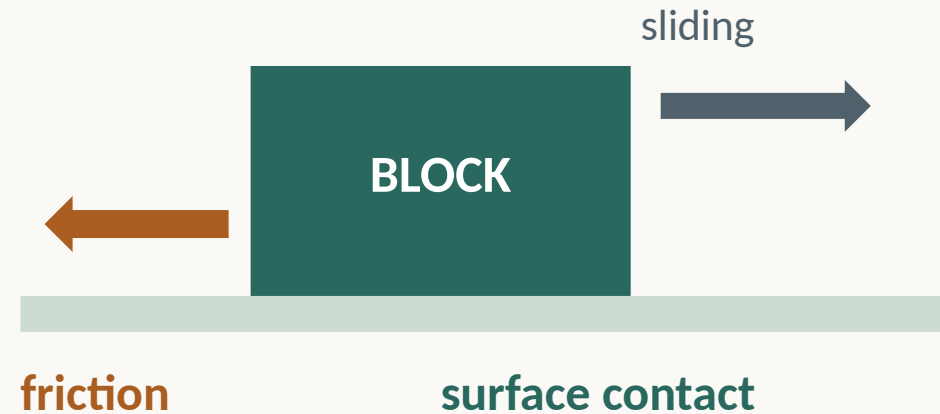
Name the two touching surfaces.

# Change one thing: the surface

Keep the block, ramp and release the same.

Change only the run-out surface.

Predict where the block will stop. Then observe.



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**What stayed the same? What changed?**

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# Helpful or unhelpful?

- 01 Bike brakes

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- 02 Trainers on a wet floor

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- 03 Moving engine parts rubbing

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- 04 Pushing a heavy box across carpet

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- 05 A door handle

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**Name the surfaces. Explain the job.**

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# The job changes the answer

## HELPS THE JOB

Braking a wheel  
Gripping the floor  
Holding a handle

## HINDERS THE JOB

Moving parts heating and wearing  
Sliding a box across carpet

# Explain both sides

**Friction is helpful here because...**

**...but it is unhelpful here because...**

Use the touching surfaces and the job in your explanation.

# Sort the claims

TRUE / FALSE / NOT A SCIENTIFIC CLAIM

- 01 Sliding friction opposes sliding at the contact.

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- 02 Friction only happens with rough surfaces.

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- 03 Oil can reduce friction in suitable lubricated contacts.

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- 04 Without friction we could not walk normally.

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- 05 Friction is a bad force.

Need a different way to take part? (optional) During the claim sort, point, say, sign or show the adult an idea, question or change that would help. You may pass or speak privately. The adult listens, explains what can change and checks whether it helped.

# Check the science

- 01 TRUE Sliding friction opposes sliding at the contact.

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- 02 FALSE Smooth surfaces can have friction too.

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- 03 TRUE Suitable lubrication can reduce friction.

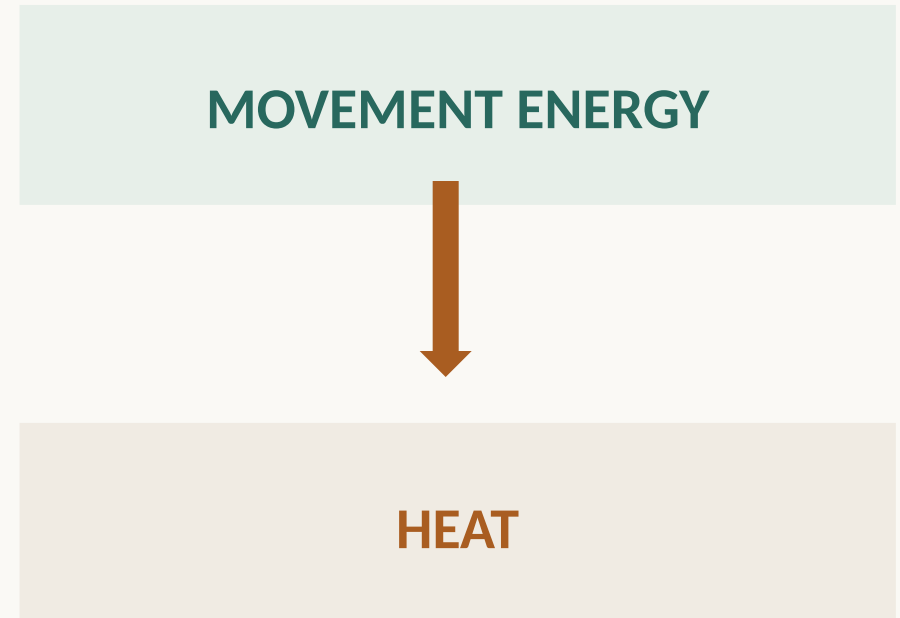
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- 04 TRUE Walking normally needs grip at the ground.

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- 05 JUDGEMENT “Bad” depends on the job.

# Build a complete explanation

**Friction is helpful at a bike brake: it transfers movement energy to heat and slows the wheel.**

It can be unhelpful between moving engine parts because rubbing heats and wears them.



# Finish Lesson A • two small checks

CHOOSE ONE ROUTE, ONCE / SUPPORTED • STANDARD • STRETCH / YOU MAY PASS

SUP 1 A box slides right on a floor. Which way does friction on the box act? Left / Right

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SUP 2 Bike brake pads rub on the wheel rim. Does this help slow the wheel? Yes / No

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STA 1 Friction acts between touching \_\_\_\_ and resists their sliding.

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STA 2 Friction at the brake pad helps the wheel \_\_\_\_.

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STR 1 Friction at a bike brake is helpful because \_\_\_\_.

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STR 2 Friction between moving engine parts can be unhelpful because \_\_\_\_.

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Point, say, sign, draw, type or use paper. An adult can scribe your words. Explanations are separate; your response is not automatically marked. Next lesson: Friction: test the surfaces.

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# Keep your prediction for next time

01

Keep your named worksheet and first prediction.

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02

Next lesson: Friction: test the surfaces.

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03

Point to one idea you want to check again.

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